

**DUY TAN UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



INDIVIDUAL PROJECT

TOPIC: Book management

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Class: CMU-CS 303 GIS

I. Class diagram:

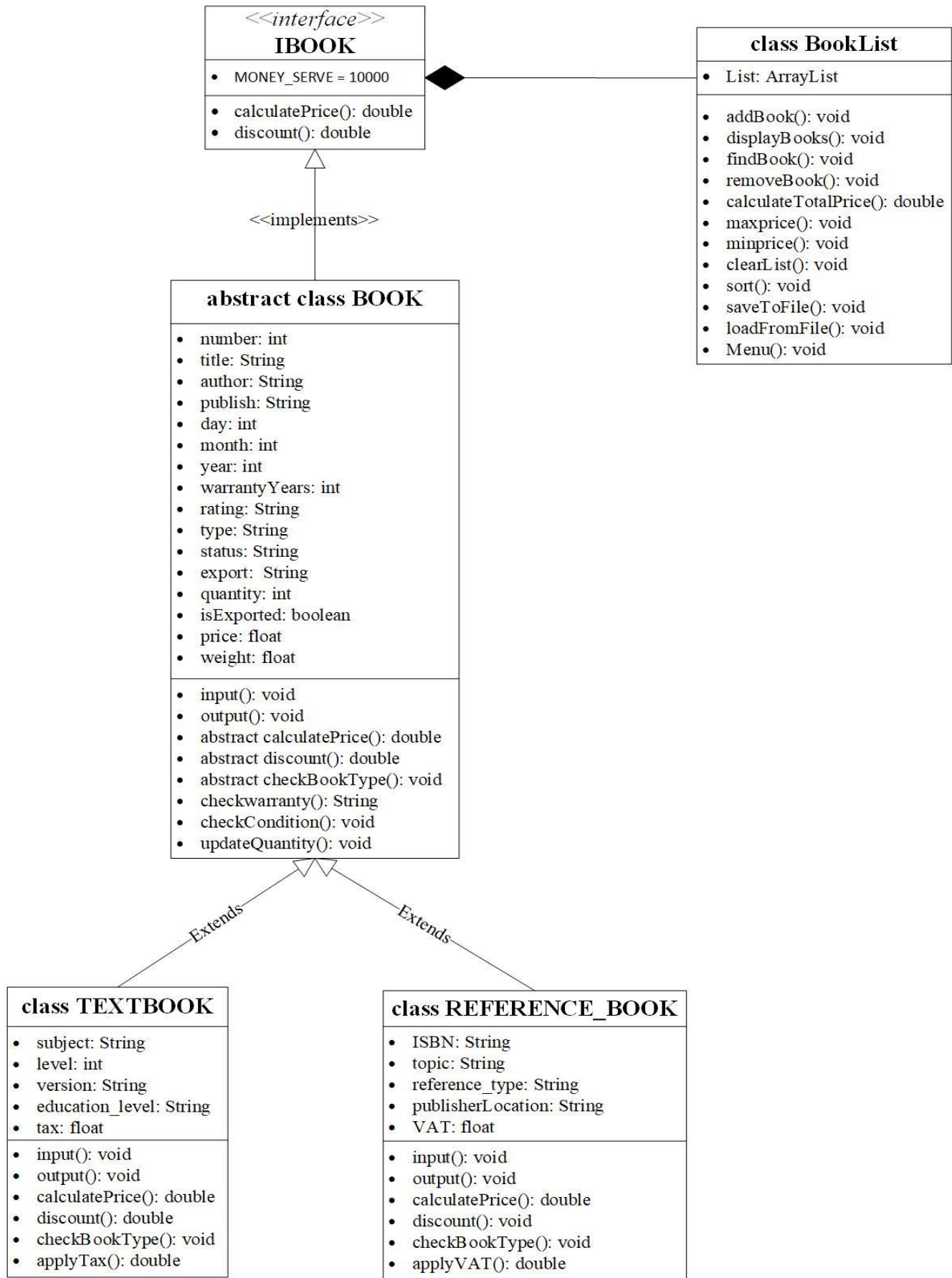


Figure 1 : Book

II. List the functions:

1. Define interface **IBOOK** with method: **calculatePrice()** and **discount()**; and with property: **MONEY_SERVE**

- **Where calculatePrice() is calculate as:**
 - ❖ With **TEXTBOOK()** = **MONEY_SERVE** + **getPrice()** * **getQuantity()** + **applyTax()** - **discount()**.
 - ❖ With **REFERECE_BOOK()** = **MONEY_SERVE** + **getPrice()** * **getQuantity()** + **applyVAT()** - **discount()**;
- **Where discount() is calculate as:**
 - ❖ With **TEXTBOOK**: **double discountRate** = 0;
 - If the status is new: **discountRate** = 0.1;
 - If the status is old: **discountRate** = 0.5;
 - Remaining: : **discountRate** = 0.05;

 - If the quantity >100: **discountRate** += 0.05;

 - If the rating is high: **discountRate** += 0.03;
 - If the rating is low: **discountRate** -= 0.02;
 - **return this.getQuantity() * this.getPrice() * discountRate;**
 - ❖ With **REFERECE_BOOK**: **double discountRate** = 0;
 - If the status is new: **discountRate** = 0.5;
 - If the status is old: **discountRate** = 1.5;
 - Remaining: : **discountRate** = 0.1;

 - If the quantity >100: **discountRate** += 2;

 - If the rating is high: **discountRate** += 0.03;
 - If the rating is low: **discountRate** -= 0.02;

 - If the reference_type is academic: **discountRate** += 0.02;
 - If the reference_type is non-academic: **discountRate** += 0.01;
 - **return this.getQuantity() * this.getPrice() * discountRate;**

2. Construct an abstract class **BOOK** implements interface **IBOOK**. It contains:

- **Properties:**
 - **int number**: Numerical order
 - **String title**: Title name
 - **String author**: Author name
 - **String publishing**: Publishing name
 - **int day**: Day of publication

- int month: Month of publication
- int year: Year of publication
- int warrantyYears: Time warranty (years)
- String rating: Product Reviews
- String type: Book genre
- String status: The condition of the product, new or old
- String export: Origin of the book
- int quantity: Number of books
- boolean isExported: Check if the book can be exported
- float price: Price of the book
- float weight: Weight of book
- **Constructor methods for:** number, title, author, publishing, day, month, year, warrantyYears, rating, type, status, export, quantity, isExported, price, weight
- **Getter methods for:** number, title, author, publishing, day, month, year, warrantyYears, rating, type, status, export, quantity, isExported, price, weight
- **Setter methods for:** number, title, author, publishing, day, month, year, warrantyYears, rating, type, status, export, quantity, isExported, price, weight
- **The methods:** input(), output(), checkwarranty(), checkCondition(), updateQuantity()
 - Create an input() method to import all the above properties into
 - Create an output() method to output all the above properties
 - Create a checkwarranty() method to check if the book is still under warranty
 - Create a checkCondition () method to check how many books are left in stock
 - Create a updateQuantity () method to add new books to the inventory
- **Abstract methods:** calculatePrice(), discount(), checkBookType()

3. Define a class TEXTBOOK extends BOOK contains:

- **Properties:**
 - String subject: Subject name
 - String level: Book level
 - String version: Published version
 - String educationLevel: Educational level of the book
 - float tax: Tax amount
- **Constructor methods for:** subject, level, version, educationLevel, tax
- **Getter methods for:** subject, level, version, educationLevel, tax
- **Setter methods for:** subject, level, version, educationLevel, tax
- **The methods:**
 - Implement the input() method of this class by calling input() from the parent class and importing new properties from this child class.
 - Implement the output() method of this class by calling output() from the parent class and exporting new properties in this child class
 - Create a calculatePrice() method to calculate the total cost of textbook
 - Create a discount() method to calculate the discount amount of the book

- Create a checkBookType() method to check the type of books that are discounted
- Create a applyTax () method to to calculate tax on books

4. Define a class REFERENCE_BOOK extends BOOK contains:

➤ **Properties:**

- String ISBN: Book code
- String topic: Topic name
- String reference_type: Name of reference book
- String publisherLocation: Place of publication
- float VAT: Enter VAT value

➤ **Constructor methods for:** : ISBN, topic, reference_type, publisherLocation, VAT

➤ **Getter methods for:** ISBN, topic, reference_type, publisherLocation, VAT

➤ **Setter methods for:** ISBN, topic, reference_type, publisherLocation, VAT

➤ **The methods:**

- Implement the input() method of this class by calling input() from the parent class and importing new properties from this child class.
- Implement the output() method of this class by calling output() from the parent class and exporting new properties in this child class
- Create a calculatePrice() method to calculate the total cost of textbook
- Create a discount() method to calculate the discount amount of the book
- Create a checkBookType() method to check the type of books that are discounted
- Create a applyVAT () method to to calculate VAT on books

5. Define a class named BOOKLIST as shown in the figure. It contains:

➤ A variable: Array that contains list of books

And perform the functions as:

- addBook(): Add new list of books
- displayBooks(): Display list of Books
- findBook(): Find books by title
- removeBook(): Delete 1 book in any position
- calculateTotalPrice(): Calculate total price of all books
- maxprice(): Find the highest priced book
- minprice(): Find the lowest priced books
- sort(): Sort list of Books ascending by calculatePrice()
- clearList(): Delete all book lists
- saveToFile(): Save list into txt file
- loadFromFile(): Read list from txt file
- The menu of program allows the user to select the functions

III. Demo results:

```
java - code - doancanhan/src/doancanhan/Doancanhan.java - Eclipse IDE
File Edit Source Refactor Source Navigate Search Project Run Window Help
Doancanhan [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (15:08:42 9 thg 10, 2024) [pid: 15288]
Choose an option:
11
Book list saved to cmutpm2.txt
Menu
1. Add Book:
2. Display Books:
3. Find Book:
4. Remove Book:
5. Calculate Total Price For Text Book:
6. Calculate Total Price For Reference Book:
7. Calculate Total Price:
8. Max Price:
9. Min Price:
10. Sort Book:
11. Save To File:
12. Read From File:
13. Clear List:
14. Exit:
Choose an option:
12
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Book Number | Title | Author | Publisher | Publication Date | Warranty Period | Rating | Type | TEXTBOOK | Status | Is Exported | Price | Weight | Subject | Level |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | do an | manh | home | 20/9/2005 | 10 | high | hoc tap | In Stock | true | 1000,00 | 50,00 | math | 2 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Book Number | Title | Author | Publisher | Publication Date | Warranty Period | Rating | Type | REFERENCE BOOK | Status | Is Exported | Price | Weight | ISBN | Topic |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 2 | code | manh | school | 30/9/2023 | 3 | high | bai tap | In Stock | true | 250,00 | 2,00 | m123 | kho |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Book list loaded from cmutpm2.txt
Menu
1. Add Book:
2. Display Books:
3. Find Book:
4. Remove Book:
5. Calculate Total Price For Text Book: |
6. Calculate Total Price For Reference Book:
7. Calculate Total Price:
8. Max Price:
9. Min Price:
10. Sort Book:
```

```
java - code - doancanhan/src/doancanhan/Doancanhan.java - Eclipse IDE
File Edit Source Refactor Source Navigate Search Project Run Window Help
Doancanhan [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (15:08:42 9 thg 10, 2024) [pid: 15288]
Enter book title:
do an
Yes!
Menu
1. Add Book:
2. Display Books:
3. Find Book:
4. Remove Book:
5. Calculate Total Price For Text Book:
6. Calculate Total Price For Reference Book:
7. Calculate Total Price:
8. Max Price:
9. Min Price:
10. Sort Book:
11. Save To File:
12. Read From File:
13. Clear List:
14. Exit:
Choose an option:
5
Total price for textbooks is: 34160.0
Menu
1. Add Book:
2. Display Books:
3. Find Book:
4. Remove Book:
5. Calculate Total Price For Text Book:
6. Calculate Total Price For Reference Book:
7. Calculate Total Price:
8. Max Price:
9. Min Price:
10. Sort Book:
11. Save To File:
12. Read From File:
13. Clear List:
14. Exit:
Choose an option:
6
Total price for reference books is: 12762.5
Menu
1. Add Book:
2. Display Books:
```

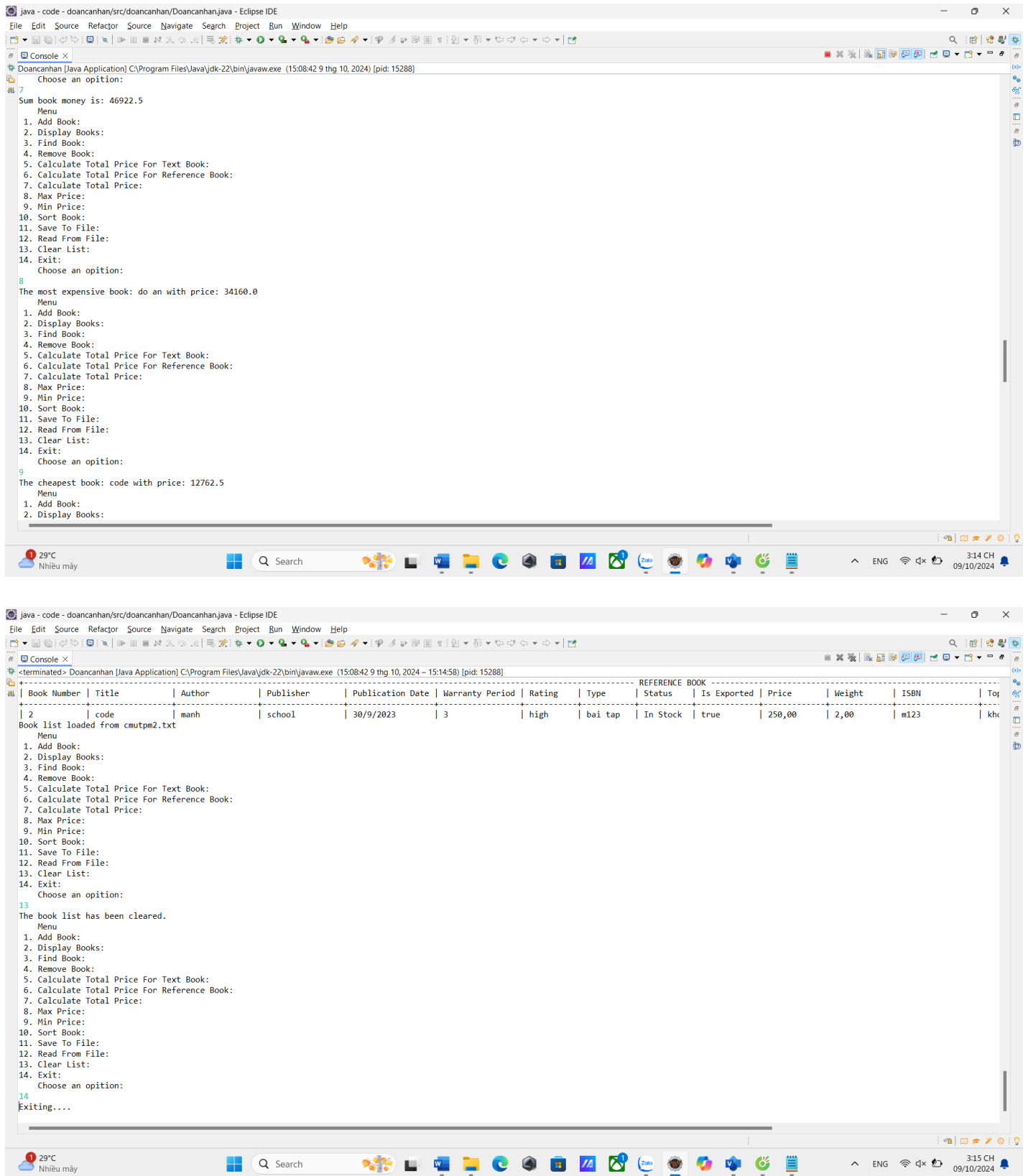


Figure 2: *Java file screen*

user if0_36587324

cmutpm2.txt

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Tệp

Chỉnh sửa

Dạng xem

TEXTBOOK																
Book Number	Title	Author	Publisher	Publication Date	Warranty Period	Rating	Type	Status	Is Exported	Price	Weight	Subject	Level	Version	Education Level	Tax
1	do an	manh	home	28/9/2005	10	high	hoc tap	In Stock	true	1000,00	50,00	math	2	new book	dai hoc	300,00
REFERENCE BOOK																
Book Number	Title	Author	Publisher	Publication Date	Warranty Period	Rating	Type	Status	Is Exported	Price	Weight	ISBN	Topic	Ref. Type	Location	VAT
2	code	manh	school	30/9/2023	3	high	bai tap	In Stock	true	250,00	2,00	m123	khoa hoc	academic	my home	200,00

Figure 3: txt file screen